Strong form

$$
\begin{aligned}
\tilde{\mathbf{v}}_r + \frac{k_0\phi_f^{1+\Theta}}{\mu_f}\nabla(\phi_f^{-1/2}\tilde{q}_f) &= 0\\
\mu_s\phi_f^{-1/2}\nabla\cdot(\phi_f^{1+\Theta}\tilde{\mathbf{v}}_r) + \frac{1}{1-\phi_f}(\tilde{q}_f - \phi_f^{1/2}q) &= 0\\
\nabla q - \nabla\cdot\hat{\sigma}(\mathbf{v}_s) &= -(1-\phi_f)\rho_r\mathbf{g}\\
\mu_s\nabla\cdot\mathbf{v}_s - \frac{\phi_f^{1/2}}{1-\phi_f}(\tilde{q}_f - \phi_f^{1/2}q) &= 0
\end{aligned}
\tag{0.1}
$$

List SI units
q $=$ pascal.
$\mathbf{v}$ $=$ m/s
Non dimensionalization choice

- $\check{\mathbf{v}} = \frac{\tilde{\mathbf{v}_r}}{k_0\rho_r g_y/\mu_f}$
- $\check{q} = \frac{q}{\rho_r g_y l_0}$
- $\check{x} = \frac{x}{l_0}$
- $l_0 = (\frac{k_0\mu_s}{\mu_f})^{1/2}$

The resulting non dimensionalized form is therefore

$$
\begin{aligned}
\check{\mathbf{v}}_r + \phi_f^{1+\Theta}\nabla(\phi_f^{-1/2}\check{q}_f) &= 0\\
\phi_f^{-1/2}\nabla\cdot(\phi_f^{1+\Theta}\check{\mathbf{v}}_r) + \frac{1}{1-\phi_f}(\check{q}_f - \phi_f^{1/2}\check{q}) &= 0\\
\nabla\check{q} - \nabla\cdot\hat{\sigma}(\check{\mathbf{v}}_s) &= -(1-\phi_f)\rho_r\hat{\mathbf{g}}\\
\nabla\cdot\check{\mathbf{v}}_s - \frac{\phi_f^{1/2}}{1-\phi_f}(\check{q}_f - \phi_f^{1/2}\check{q}) &= 0
\end{aligned}
\tag{0.2}
$$

scaled nondimensionalized weak form let $\check{\sigma} = 2(\mathcal{D}\mathbf{v} - \frac{1}{3}\nabla\cdot\mathbf{v}\mathbf{I})$

$$
\begin{aligned}
(2\phi_s\check{\sigma}(\mathbf{v}_{s,h}), \nabla\Psi_s) - (\check{q}_h, \nabla\cdot\Psi_s) &= -(\phi_s\check{\mathbf{f}}, \Psi_s)\\
(\nabla\cdot\check{\mathbf{v}}_{s,h}, \omega) + \left(\frac{\hat{\phi}_f}{\phi_s}\check{q}_h, \omega\right) - \left(\frac{\hat{\phi}_f^{1/2}}{\phi_s}\check{q}_{f,h}, \omega\right) &= 0\\
(\check{\mathbf{v}}_{r,h}, \Psi_r) - (\check{q}_{f,h}, \hat{\phi}_f^{-1/2}\nabla\cdot(\phi_f^{1+\Theta}\Psi_r)) &= 0\\
(\hat{\phi}_f^{-1/2}\nabla\cdot(\phi_f^{1+\Theta}\Psi_r), \omega_f) + \left(\frac{1}{\phi_s}\check{q}_{f,h}, \omega_f\right) - \left(\frac{\hat{\phi}_f^{1/2}}{\phi_s}\check{q}_h, \omega_f\right) &= 0
\end{aligned}
\tag{0.3}
$$